

Semantic Tech Labs

Building Intelligent Systems with Large Language Models and Autonomous Agents

Company Profile • Version 2.4 • November 2024

1. About the Company

Semantic Tech Labs is an artificial intelligence product and research company headquartered in Hyderabad, India. Founded in 2022 by **Bojja Akash**, the company builds AI tools and automation platforms that are centered around Large Language Models (LLMs) and autonomous agents. Our mission is to make intelligent automation accessible to businesses of every size, from early-stage startups to large enterprises.

At Semantic Tech Labs, we believe that the next decade of software will be defined by systems that can reason, plan, and act. Our products allow organizations to deploy LLM-powered agents that handle real-world workflows such as customer support, document analysis, research summarization, and process automation without requiring deep in-house ML expertise.

1.1 Mission

To build trustworthy AI agents that augment human teams, reduce operational toil, and unlock new categories of products powered by large language models.

1.2 Vision

A world where every business has an always-on, reliable AI workforce — capable of understanding context, taking actions, and continuously learning from feedback.

2. Founder Profile

Bojja Akash is the founder and Chief Executive Officer of Semantic Tech Labs. Akash holds a Bachelor's degree in Computer Science from the Indian Institute of Technology

(IIT) Hyderabad and previously worked on applied machine learning teams in the fin-tech and SaaS industries. He started Semantic Tech Labs after observing that most companies struggled not with building LLM prototypes, but with making them reliable enough for production workloads.

Under his leadership, the company has grown from a two-person team to a 48-person organization with clients across India, the United Arab Emirates, Singapore, and the United States. Akash continues to lead product strategy and the company's open-source initiatives in the agent orchestration space.

3. Products & Services

Semantic Tech Labs develops a portfolio of AI tools that share a common agent runtime. Each product is designed to be modular, observable, and safe to deploy in regulated industries. Our flagship offerings include:

3.1 AgentFlow

AgentFlow is a visual platform for designing, deploying, and monitoring multi-step LLM agents. Users can chain tools, retrieval pipelines, and human-in-the-loop checkpoints without writing infrastructure code. AgentFlow ships with more than 60 pre-built integrations including Slack, Gmail, Salesforce, Notion, Jira, and PostgreSQL.

3.2 SemantiQA

SemantiQA is a retrieval-augmented generation (RAG) platform aimed at enterprise knowledge bases. It supports hybrid search, document-level access controls, citation tracking, and evaluation harnesses that measure answer faithfulness against source documents. SemantiQA is used by legal, healthcare, and financial services customers to ground LLM responses in their private corpora.

3.3 InsightLoop Analytics

InsightLoop is an analytics automation product that connects to business intelligence warehouses and generates narrative reports, anomaly explanations, and recommended actions using LLM agents. It is positioned as a virtual analyst for revenue, marketing, and operations teams.

4. Technology Stack

Our platform is model-agnostic and supports multiple foundation model providers so that customers are never locked into a single vendor. The core stack includes:

- **Foundation Models:** OpenAI GPT-4o, Anthropic Claude 3.5 Sonnet, Google Gemini 1.5 Pro, Meta Llama 3.1 70B (self-hosted), and Mistral Large.
- **Embeddings:** OpenAI text-embedding-3-large, BGE-M3, and Cohere embed-multilingual-v3.
- **Vector Databases:** Pinecone, Weaviate, Qdrant, and pgvector for Postgres deployments.
- **Orchestration:** LangGraph for agent state machines, custom DAG executor for tool calls.
- **Backend:** Python 3.11, FastAPI, Celery, Redis, and PostgreSQL 16.
- **Frontend:** React 18, TypeScript, Next.js 14, and TailwindCSS.
- **Infrastructure:** Kubernetes on AWS (EKS) with Terraform-managed environments, and GPU nodes powered by NVIDIA A10G and H100 for self-hosted inference.

5. Office Locations

Semantic Tech Labs operates from a primary headquarters in Hyderabad and a smaller engineering satellite office in Bengaluru. The registered corporate address is:

Location Type	Address
Registered Office	Plot 42, Hi-Tech City Road, Madhapur, Hyderabad, Telangana 500081, India
Engineering Office	Tower B, Knowledge City, Raidurg, Hyderabad, Telangana 500032, India
Satellite Office	4th Floor, Prestige Tech Park, Kadubeesanahalli, Bengaluru, Karnataka 560103, India
Phone	+91 40 4892 3310
Email	hello@semantictchlabs.ai

6. Company Metrics (FY 2023–2024)

Metric	FY 2023	FY 2024
Annual Recurring Revenue (USD)	\$1.4M	\$4.7M
Active Enterprise Customers	23	61
Monthly Agent Executions	8.2M	44.6M
Tokens Processed (Billions)	3.1	17.9
Employee Headcount (Year-end)	19	48

© 2024 Semantic Tech Labs Pvt. Ltd. All rights reserved. Confidential.